Reliability-Oriented Principal Component Analysis in StatCore: Exact Algorithms and Empirical Limits

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***Abstract*—This paper documents the pure-NumPy StatCore PCA family by executable behavior rather than names alone. We define base SVD PCA and all selectable variants: supervised row-aware weighted PCA, residual-reweighted robust PCA, thresholded sparse power iterations with deflation, and a closed-form probabilistic PCA (PPCA) fit. Centering, weights, stopping, transformation, reconstruction, assumptions, and complexity are specified exactly. The robust and sparse methods are heuristics without global guarantees; the PPCA inverse transform also differs algebraically from conventional conditional-mean reconstruction. Five-seed experiments on contaminated synthetic data and bundled Wine compare held-out reconstruction MSE against scikit-learn PCA as the standard reference. Results are measured rather than inferred, and do not establish general superiority.**

***Index Terms*—principal component analysis, robust statistics, sparse PCA, probabilistic PCA, reliability, NumPy**

I. Introduction

PCA chooses a low-dimensional linear subspace minimizing squared Euclidean reconstruction error, whose classical solution follows from the singular value decomposition (SVD) [1], [2]. StatCore provides this reference and four alternatives motivated by reliability. Since terms such as "robust" and "sparse" cover many algorithms, we specify the behavior of `src/ml_reliability/statcore/pca/pca.py`, not an idealized textbook substitute. The API requires finite numeric $X \in \mathbb{R}^{n\times d}$, at least two rows and one feature; k must be integer in $[1, \min(n, d)]$. Optional y must be a one-dimensional sample-aligned vector. Features are not automatically standardized.

II. Notation and standard PCA

Let rows be $x_i^\top$, mean $\bar{x} = n^{-1}\sum_i x_i$, centered matrix $X_c = X - \mathbf{1}\bar{x}^\top$, and thin SVD $X_c = U\Sigma V^\top$, with singular values $s_1 \geq \cdots \geq 0$. For orthonormal row loadings $C \in \mathbb{R}^{k\times d}$,

$$\min_{CC^\top = I_k} \|X_c - X_c C^\top C\|_F^2. \tag{1}$$

Expanding gives $\|X_c\|_F^2 - \operatorname{tr}(CX_c^\top X_c C^\top)$; the Ky Fan maximum principle yields $C = V_k^\top$, minimum $\sum_{j>k} s_j^2$, with nonunique optimal subspace under tied eigenvalues. The `base` implementation uses this centered thin SVD. It reports population explained variances s_j^2/n, not unbiased sample variances. Scores are $X_c C^+$ (equal to $X_c C^\top$ for orthonormal C); inverse transform is $ZC + \bar{x}$. Zero singular values yield zero variance without invalidating the SVD basis. Dense SVD costs $O(nd\min(n, d))$.

III. Implemented variants

A. *supervised_row_aware: adaptive weighted PCA*

An ordinary centered SVD first supplies the leading $q = \min(k, \operatorname{rows}(V^\top))$ loadings. Scores $t_i = (x_i - \bar{x})V_q$ and variances $v_j = \max(s_j^2/n, 10^{-12})$ define $D_i^2 = \sum_j t_{ij}^2/v_j$. Let $h = 2.5\,\mathrm{median}(D^2) + 10^{-12}$. Initial outlier factor is $a_i = \min(1, h/D_i^2)$ for $D_i^2 > h$, and one otherwise. If labels are given, multiply by $n/(C_y n_{y_i})$, with C_y the number of observed classes and n_c their counts. Normalize final weights to $w_i = na_i/\sum_j a_j$, so $\sum_i w_i = n$. The weighted mean is $\mu_w = \sum_i w_i x_i/n$. Components are the top right singular vectors of $\operatorname{diag}(\sqrt{w})(X - \mathbf{1}\mu_w^\top)$. Conditional on these data-derived weights, this exactly minimizes $\sum_i w_i\|(x_i - \mu_w) - (x_i - \mu_w)C^\top C\|^2$ over orthonormal loadings: the ordinary SVD trace derivation applies to the weighted matrix. Labels enter only through inverse class frequency; this is not a discriminant objective. Class rebalancing may amplify noisy

minority examples. Weights are positive for finite data; the exact weighted PCA solution has no iterative convergence issue. The extra initial SVD and weighted SVD each cost $O(nd\min(n,d))$. Reported variance is $s_j^2/\max(\sum_i w_i, 1)$, while explained-variance ratios divide by the unweighted total centered variance, and therefore need not sum to one.

B. robust: median start and residual reweighting

The coordinatewise median m centers the data, $A = X - \mathbf{1}m^\top$. Coordinate scales are $b_j = 1.4826\,\mathrm{median}\,|A_{ij}| + 10^{-12}$; row magnitudes $r_i = \|A_i/b\|_2$ give initial cutoff $c_0 = \mathrm{median}(r) + 1.345(1.4826)\,\mathrm{median}\,|r - \mathrm{median}(r)|$. Initial weights are $w_i = [\min(1, c_0/\max(r_i, 10^{-12}))]^2$. The initial exponent is two; subsequent reweighting is not squared. For at most ten rounds, compute weighted location $\mu_t = \sum_i w_i A_i / \sum_i w_i$, weighted SVD of $A - \mathbf{1}\mu_t^\top$, and top k loadings C_t. Orthogonal residual $e_i = (A_i - \mu_t) - (A_i - \mu_t)C_t^\top C_t$, norm $q_i = \|e_i\|$, scale $s = 1.4826\,\mathrm{median}\,|q - \mathrm{median}(q)| + 10^{-12}$, and cutoff $c = \mathrm{median}(q) + 1.345s$ define $w_i^{new} = \min(1, c/\max(q_i, 10^{-12}))$. Stop when $\max_i |w_i^{new} - w_i| < 10^{-6}$, or after ten iterations. The public mean remains the coordinate median; internal weighted means affect the fitted directions only. This is a Huber-motivated reweighting heuristic [3], not an optimizer of one fixed robust PCA objective. Its initialization uses standardized coordinatewise outlyingness; later weights use orthogonal residuals, and both location and subspace change. No monotonic decrease, stationarity, or breakdown-point guarantee follows. Maximum cost is ten dense weighted SVDs, about $O(10nd\min(n,d))$. Epsilon guards constant coordinates/zero residuals. The final weights can be updated after the tenth SVD and thus not exactly match its fitted components. A robust fit may suppress legitimate rare directions. **Algorithm 1: Implemented robust reweighting.**

1) Set location to the coordinate median; calculate scaled row magnitudes and squared initial clipped weights as defined above.
2) For at most ten rounds, compute the weighted mean and top-k right singular vectors; form orthogonal residual norms.
3) Re-estimate the MAD scale and cutoff; set each new weight to $\min(1, c/\max(q_i, 10^{-12}))$.
4) Stop if the maximum weight change is below 10^{-6}; otherwise continue. Return the last SVD directions, median location, and current weights.

C. sparse: thresholded power iterations and deflation

Starting with $R = X_c$, each of k components initializes v as the leading right singular vector of R plus $10^{-3}\xi$, $\xi \sim N(0, I_d)$, seeded by NumPy `default_rng(random_state)`; it normalizes v. Up to `n_iter` (default 100) iterations compute

$$u = Rv/\|Rv\|_2, \qquad z = R^\top u,$$
$$\tau = \alpha\|z\|_\infty, \qquad \tilde{v} = \mathrm{sign}(z)(|z| - \tau)_+,$$
$$v_{new} = \tilde{v}/\|\tilde{v}\|_2.$$

If $\|Rv\| < 10^{-12}$, stop; if $\|\tilde{v}\| < 10^{-12}$, use z instead; stop on $\|v_{new} - v\| < 10^{-8}$. The loading is appended and $R \leftarrow R - (Rv)v^\top$. This sequential rank-one deflation does not enforce orthogonality. The threshold is a fraction α (default 0.1) of the maximum absolute correlation, not an absolute penalty parameter. It resembles sparse power methods but is not an exact global solution to a canonical sparse-PCA objective, and no global monotonic guarantee is available. Cost is up to $O(knd\,\texttt{n_iter})$ multiplies plus a leading SVD per component, up to $O(knd\min(n,d))$. Stored loadings C can be nonorthogonal. Transform uses X_cC^+; inverse is $ZC + \bar{x}$, i.e., least-squares reconstruction in the loading span. The reported variance is coordinatewise population variance of X_cC^+, not a variance decomposition or the thresholded objective. Defaults are $k = 2, \alpha = .1, \texttt{n_iter} = 100, \texttt{random_state} = 0$; ranges are not validated. Degenerate residuals can trigger zero-vector fallbacks.

D. probabilistic: closed-form PPCA fit

PPCA models $x = Wz + \mu + \epsilon$, $z \sim N(0, I_k)$, $\epsilon \sim N(0, \sigma^2 I_d)$. For centered SVD eigenvalues $\lambda_j = s_j^2/n$, the maximum-likelihood covariance solution for Gaussian iid observations uses leading eigenvectors and $\hat{\sigma}^2 = (d-k)^{-1}\sum_{j>k}\lambda_j$ when $d > k$, $\hat{W}_j = v_j\sqrt{\max(\lambda_j - \hat{\sigma}^2, 0)}$ [4]. Code floors noise variance at 10^{-10}, uses zero before flooring if $d \le k$, stores leading $V_k^\top$, and reports $\lambda_1, \ldots, \lambda_k$. No likelihood optimization is run. With $M = W^\top W + \sigma^2 I$, conventional posterior mean is $M^{-1}W^\top(x - \mu)$. The implemented transform $(x - \mu)^\top W M^{-\top}$ agrees since M is symmetric. But inverse transform is $z^\top(MW^+) + \mu$, not conventional PPCA conditional-mean reconstruction $z^\top W^\top + \mu$. Even for nonzero orthogonal columns this introduces eigenvalue-dependent rescaling; zero columns accentuate the discrepancy. This code-specific round trip is not generally the PPCA conditional expectation. The fitted covariance model is conventional; the API decoder is not.

E. Shared API, edge conditions, and comparison

All methods require finite arrays and fixed inference feature count. `score_samples` is per-row MSE using each method's actual encode/decode path; `evaluate` returns reconstruction metrics. There is no missing-value handling, whitening, automatic scaling, or uncertainty calibration. Ratios use the unweighted total variance and return zeros for zero-total variance. Components are orthogonal except sparse; methods have different score conventions.

IV. Empirical Study

The attached `experiments/pca_experiment.py`, run from repository root using the project interpreter, compares StatCore and scikit-learn `PCA` as the ordinary reference [5]. Synthetic data use a two-dimensional latent Gaussian embedded in six dimensions; 320 contaminated training rows include 24 featurewise gross outliers while 80 test rows are clean. Wine is the bundled 178-row, 13-feature classification dataset; each seed creates a deterministic 75/25 split, with

TABLE I
IMPLEMENTED PCA VARIANTS: EXACT OPERATION AND WARRANTED GUARANTEE.

Variant	Location and weighting	Direction estimator	Guarantees / caveats	Controls (defaults)	Complexity
base	Arithmetic mean, unit weights	Top right SVD vectors	Global squared-error optimum	$k = 2$	One SVD
supervised row-aware	Weighted mean; outlier and optional class-balance weights	Weighted SVD	Exact PCA conditional on weights; labels do not enter a discriminant loss	optional y; cutoff 2.5	Two SVDs
robust	Coordinate median; initial squared then residual Huber weights	Up to 10 weighted SVD re-fits	Heuristic, no common monotone objective	cutoff 1.345; stop 10^{-6}	Up to 10 SVDs
sparse	Arithmetic mean, unit weights	Soft-thresholded alternating powers and deflation	No global optimum; loadings not orthogonal	$\alpha = .1$, 100 rounds, seed 0	Up to k SVDs + iterations
probabilistic	Arithmetic mean; isotropic noise ML estimate	Leading eigenvectors and PPCA loading scales	Gaussian covariance fit; noise floor inverse map nonstandard	10^{-10}	One SVD

TABLE II
HELD-OUT RECONSTRUCTION MSE, MEAN $\pm$ SAMPLE SD OVER FIVE SEEDS (LOWER IS BETTER).

Method	Synthetic	Wine
Reference PCA	0.0749 ± 0.0387	0.5055 ± 0.0292
base	0.0749 ± 0.0387	0.5055 ± 0.0292
supervised_row_aware	0.0067 ± 0.0058	0.5027 ± 0.0299
robust	0.0021 ± 0.0002	0.5195 ± 0.0332
sparse	0.0755 ± 0.0420	0.5101 ± 0.0304
probabilistic	0.0749 ± 0.0387	0.5055 ± 0.0292

labels supplied only to the row-aware method. All methods share training-only standardization (zero scales replaced by one), $k = 2$, seeds 11, 23, 37, 51, 71. Metric is held-out reconstruction MSE, lower better; entries are mean and sample SD. The script writes numeric LaTeX cells only after measurement. These are small descriptive experiments, not claims of statistical significance. Standardization changes the PCA estimand; the synthetic contamination is controlled; Wine is not a deployment population; and reconstruction error is not downstream prediction quality or calibrated uncertainty. Scikit-learn is used only for the reference estimator and dataset loading. A method may be worse than the reference, and all comparisons require task-specific validation.

V. CONCLUSION

Base PCA has the classical optimum; row-aware weighted SVD exactly solves its conditional weighted problem; robust and sparse methods are finite heuristics whose guarantees must not be overstated. PPCA's covariance fit follows the standard closed form while its inverse transform is implementation-specific and nonstandard. The measured comparisons expose trade-offs without asserting universal gains.

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